

Deep Reference Priors: What is the best way to pretrain a model?

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Problem Statement

$$\hat{P}_n = \{(x_i, y_i)\}_{i=1}^n; x_i \in \mathbb{R}^d; y_i \in \{1 \dots C\}$$

$$x^n = (x_1, \dots x_n); y^n = (y_1 \dots y_n)$$

$$w \in \mathbb{R}^p; p_w(y|x)$$

$$z = y|x; \pi(w) - \textit{prior}$$

$$\pi^* = \operatorname{argmax}_{\pi} l_{\pi}(w; z)$$

Problem Statement

Key Question

What is the optimal way to exploit extra data (labeled/unlabeled, from same/related tasks) to pretrain a model for a given task?

Core Idea

A pretrained representation can be viewed as a **Bayesian prior**. The optimal prior should allow the target task data to maximally influence the posterior.

Formalization

Use **Reference Priors** theory: priors that maximize mutual information between model weights and data:

$$\pi^* = \operatorname{argmax}_{\pi} I_{\pi}(w; z)$$

Reference Priors vs Jeffreys Prior

Jeffreys Prior

- ▶ Goal: **Reparameterization invariance**
- ▶ Defined via Fisher Information Matrix: $\pi_J(w) \propto \sqrt{\det g(w)}$
- ▶ **Independent** of sample size n
- ▶ Always **continuous**

Reference Prior

- ▶ Goal: **Maximize data influence** on posterior
- ▶ Maximizes mutual information: $\pi^* = \arg \max I_{\pi}(w; z)$
- ▶ **Depends** on sample size n (adaptive)
- ▶ **Discrete** for finite n (supported on atoms)

Reference priors

In fact we want to maximize $KL(p(w|z)|\pi(w))$

$$\pi^* = \operatorname{argmax}_{\pi} I_{\pi}(w; z) = \int p(z)p(w|z) \log \left[\frac{p(w|z)}{\pi(w)} \right] dz dw = H(w) - H(w|z)$$

Where $p(z) = \int \pi(w)p(z|w)dw$ and $H(w) = -\mathbb{E}_w[\log \pi(w)]$

It is difficult to compute reference prior straight-through so we assume that $\pi_n^* = \operatorname{argmax}_{\pi} I(w; z^n) = H(w) - H(w|z)$

$$\lim_{n \rightarrow \infty} \pi_n^* = \pi^*$$

NOTE! reference prior does not exist if $I_{\pi}(w; z^n)$ is infinite

Blahut-Arimoto Algorithm

Iterations

$$\pi^{t+1}(w) \propto \exp(KL(p(z|w)|p(z)))\pi^t(w)$$

Dependence on Sample Size

- ▶ Small n : Prior concentrates on few diverse models
- ▶ Large n : Prior converges to Jeffreys prior
- ▶ Enables adaptation to data scarcity

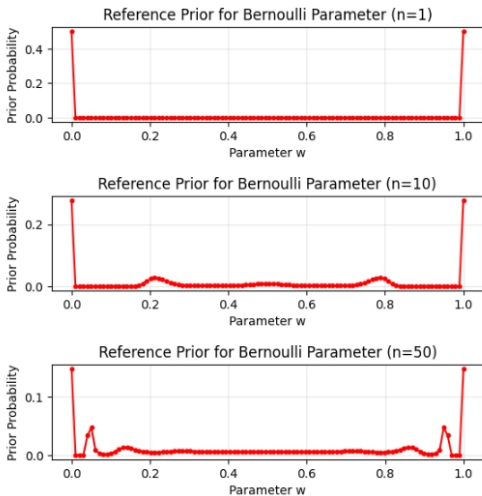
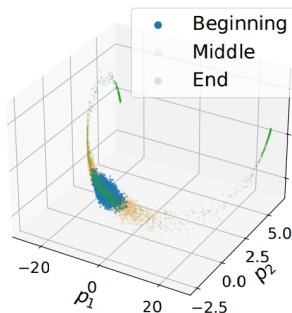


Figure: Reference prior for coin-tossing ($n=1,10,50$)

Computing Reference Priors for Deep Networks

Blahut-Arimoto Algorithm with Particles

- ▶ Represent prior as set of particles: $\{w^1, \dots, w^K\}$
- ▶ Each particle is a neural network
- ▶ Gradient of mutual information flows to all particles
- ▶ Particles become diverse in prediction space



Application 1: Semi-Supervised Learning

Formalization

Use unlabeled data x^u to build optimal prior $\pi^*(w)$:

$$\max_{\pi} \gamma l_{\pi}(w; y^u, x^u) + \mathbb{E}_{w \sim \pi} [\log p(y^n | x^n, w)]$$

Connection to Practice

- ▶ Explains **consistency regularization** (FixMatch)
- ▶ Explains **entropy minimization** techniques
- ▶ Provides theoretical foundation for SSL

Practical Implementation

- ▶ Heavy data augmentation (RandAugment, CTAugment)
- ▶ Jensen's inequality bound for entropy computation
- ▶ Pseudo-labeling with confidence threshold

Application 2: Transfer Learning

Two-Stage Experiment

- ▶ Stage 1: Source task data (x_s^m, y_s^m)
- ▶ Stage 2: Target task data (x_t^n, y_t^n)
- ▶ Goal: Optimal prior for target task

Optimal Prior Formula

$$\pi_{n|m}^\beta(w) \propto \pi_n^*(w) \cdot p(y_s^m | x_s^m, w)^{-\beta}$$

- ▶ $\beta = 1$: Forget source task (different tasks)
- ▶ $\beta = 0$: Remember source task (similar tasks)
- ▶ Connects to **Predictive Information Bottleneck**

Experimental Results: Semi-Supervised Learning

CIFAR-10 Classification Accuracy (%)

Method	50 samples	250 samples	500 samples
FixMatch (RA)	86.19	94.93	93.91
FlexMatch	95.03	95.02	-
Deep Reference Prior	85.45	92.13	92.94

Key Findings

- ▶ Competitive with state-of-the-art methods
- ▶ Particularly effective with very few labeled samples (5 per class)
- ▶ Provides theoretical explanation for empirical successes

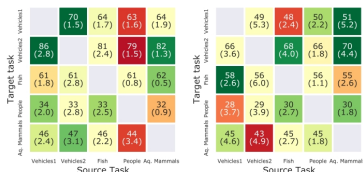
Experimental Results: Transfer Learning

Setup

- ▶ CIFAR-100, 20 five-way classification tasks
- ▶ Source: 1000 labeled samples
- ▶ Target: 100 labeled samples + unlabeled data

Results

- ▶ **Outperforms fine-tuning** in low-data regime
- ▶ Effectively leverages both source labels and target unlabeled data
- ▶ Adaptive to task relatedness via β parameter



Conclusions

- ▶ possible to use in deep neural networks
- ▶ maintains the reparametrization invariance of Jeffreys prior
- ▶ helps if the number of samples from the task available to the learner is finite
- ▶ uses sample size while Jeffrey's prior doesn't do it